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Possible mechanisms of sugar maple regeneration failure and replacement by beech in the Boisé-des-Muir old-growth forest, Québec¹

Marilou BEAUDET², Christian MESSIER & David PARÉ³, Groupe de recherche en écologie forestière interuniversitaire (GREFI), Département des sciences biologiques, Université du Québec à Montréal, C.P. 8888, Succ. Centre-Ville, Montréal, Québec, H3C 3P8, Canada.

Jacques BRISSON, Institut de recherche en biologie végétale, 4101 rue Sherbrooke Est, Montréal, Québec, H1X 2B2, Canada.

Yves BERGERON, Groupe de recherche en écologie forestière interuniversitaire (GREFI), Département des sciences biologiques, Université du Québec à Montréal, C.P. 8888, Succ. Centre-Ville, Montréal, Québec, H3C 3P8, Canada.

Abstract: To better understand the causes of sugar maple (*Acer saccharum* Marsh.) replacement by beech (*Fagus grandifolia* Ehrh.) in the Boisé-des-Muir old-growth forest (Québec), we studied, *i*) the effect of beech litter on soil properties, *ii*) the effect of low light availability on maple survival, and *iii*) beech regeneration via root sprouts. Soil properties, understory light, seedling emergence and survival, height growth, and density distribution of maple and beech were compared between the old-growth stand and an adjacent managed stand where the abundance of maple regeneration was high. We did not observe any detrimental effect of beech litter on soil properties and maple early establishment. Overall light availability and maple stocking were lower, and maple density declined more markedly with increasing size in the old-growth stand than in the managed stand. The deficient maple regeneration in the old-growth stand might be related to a lack of microsites where light availability is high enough for maple survival, and not to a lack of initial recruitment in this species. In the old-growth stand, beech was located in more shaded microsites than maple but had a higher growth. Size class distributions of maple and beech suggest that beech had a higher survival rate than maple and that beech sprouts had a higher survival rate than beech seedlings. Beech seems to be better adapted than maple to survive in the deep shade of the old-growth stand. However, we expect that maple will regain its dominance in the understory of the Boisé-des-Muir with increasing stand breakup that is likely to occur in the future.

Keywords: *Acer saccharum*, *Fagus grandifolia*, regeneration, soil properties, understory light availability, root sprouting, old-growth forest, succession.

Résumé: Afin de mieux comprendre les causes du remplacement de l'érable à sucre (*Acer saccharum* Marsh.) par le hêtre (*Fagus grandifolia* Ehrh.) dans la forêt ancienne du Boisé-des-Muir (Québec), nous avons étudié: *i*) l'effet de la litière de hêtre sur les caractéristiques du sol; *ii*) l'effet de la faible disponibilité en lumière sur la survie de l'érable; et *iii*) la reproduction du hêtre par drageonnement. Les caractéristiques du sol, la lumière, l'émergence et la survie des semis, la croissance en hauteur et la densité de la régénération d'érable et de hêtre ont été comparées entre la forêt ancienne et une forêt aménagée adjacente, où l'abondance de la régénération d'érable à sucre était plus élevée. Nous n'avons pas observé d'effet nocif de la litière de hêtre sur les caractéristiques du sol et sur l'établissement de l'érable. La disponibilité en lumière moyenne et le coefficient de distribution de l'érable étaient plus bas, et la densité des érables diminuait plus rapidement avec l'augmentation de la taille dans la forêt ancienne comparativement au peuplement aménagé. La faible régénération d'érable dans la forêt ancienne pourrait être attribuable au nombre restreint de microsites où la lumière est suffisante pour permettre la survie de cette espèce, et non pas à une déficience au niveau de son établissement initial. Dans la forêt ancienne, le hêtre était situé dans des microsites plus ombragés que l'érable mais avait une croissance plus forte. La comparaison de la densité par classe de taille des érables et des hêtres suggère que le hêtre aurait un taux de survie plus élevé que l'érable et que les drageons de hêtre auraient une meilleure survie que les semis. Le hêtre semble être mieux adapté que l'érable pour survivre dans les conditions très ombragées du Boisé-des-Muir. Toutefois, nous nous attendons à ce que l'érable retrouve sa dominance dans le sous-bois du Boisé-des-Muir à la suite des perturbations de plus grande envergure qui surviendront probablement dans le futur.

Mots-clés: *Acer saccharum*, *Fagus grandifolia*, régénération, caractéristiques du sol, disponibilité en lumière en sous-bois, drageonnement, forêt ancienne, succession.

Introduction

The sugar maple-American beech (*Acer saccharum* Marsh.-*Fagus grandifolia* Ehrh.) forest type is widespread in eastern North America and its dynamics has been the subject of numerous studies. Much effort has been invested in documenting and explaining patterns of tree replacement

or coexistence between maple and beech (Forcier, 1975; Woods, 1979; Cypher & Boucher, 1982; Leak, 1987; Canham, 1989; Runkle, 1990; Brisson, Bergeron & Bouchard, 1992; Brisson *et al.*, 1994; Kupfer & Runkle, 1996; Poulson & Platt, 1996).

The Boisé-des-Muir Ecological Reserve is an 11 ha old-growth sugar maple-beech stand located at the northern limit of the temperate deciduous forest. Brisson *et al.* (1994) showed that the species composition of this stand is not at

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²Corresponding author, e-mail: d160474@er.uqam.ca or marilou.beaudet@sympatico.ca

³Alternate address: Ressources naturelles Canada, Service canadien des forêts, 1055 rue du PEPS, C.P. 3800, Sainte-Foy, Québec, Canada, G1V 4C7.

equilibrium, and that beech is increasing in abundance over time at the expense of sugar maple. The conclusions of Brisson *et al.* (1994) come from predictions of future tree composition from transition matrix models based on actual understory composition. Their results indicated that the current number of sugar maple saplings is not sufficient for the maintenance of the present tree species composition. The reasons for this shift in composition are still not well understood, but Brisson *et al.* (1994) suggested three hypotheses.

First, the observed deficiency in sugar maple regeneration could be related to changes in soil properties resulting from the increasing proportion of beech in the forest stand. Beech litter has a high C:N ratio (Bouliane, 1962), a low nutrient concentration (Côté & Fyles, 1994), a low pH (Bouliane, 1962), and its decomposition rate is relatively slow (Melilo, Aber & Muratore, 1982).

Second, shade-induced stress in the understory of the old-growth stand could be more detrimental to sugar maple than to beech regeneration (Brisson *et al.*, 1994). Canham *et al.* (1994) have shown that beech and hemlock canopies have the strongest light attenuation properties among several tree species of eastern North America. Although sugar maple is known to be very shade tolerant, mortality of sugar maple is higher than that of beech under very shady conditions (Canham, 1988; Pacala *et al.*, 1994; Kobe *et al.*, 1995; Poulson & Platt, 1996). Therefore, low understory light availability resulting from the presence of beech in the canopy could impair sugar maple establishment, growth, and survival, and favor beech regeneration.

Third, Brisson *et al.* (1994) found that at the Boisé-des-Muir many beech saplings were root sprouts, and suggested that the ability of beech to reproduce vegetatively could favor beech survival in the shady understory. Root sprouts differ from individuals of seed origin in many ways. Root sprouts do not need to allocate resources to the construction of a root system, and could benefit from parental subsidies coming from a well-illuminated parent tree. Ward (1961) observed higher radial growth and better resistance to browsing and insect defoliation in sprouts than in individuals of seed origin. Also, Kobe *et al.* (1995) observed that beech had higher survival than eight other species including sugar maple, and suggested that this could be due to the fact that the beech saplings under study were predominantly root sprouts. Although many researchers have questioned the ability of root sprouts to develop into canopy trees (Woods, 1979; 1984; Tubbs & Houston, 1990; Poulson & Platt, 1996), in some geographic areas there is no doubt concerning the importance of root sprouting in the self-replacement of beech (Ward, 1961; Jones & Raynal, 1986).

The objective of this study was to investigate the possible contribution of the three mechanisms described above in explaining the gradual replacement of sugar maple by beech in the Boisé-des-Muir old-growth forest, Québec. In order to do so, soil conditions, light availability, seedling emergence and survival, height growth, and density distribution of maple and beech regeneration were compared between the Boisé-des-Muir old-growth stand, where the number of maple saplings is low, and an adjoining managed maple stand, where the number of maple saplings is high.

Material and methods

STUDY SITE

The Boisé-des-Muir Ecological Reserve is located near Huntingdon, 70 km southwest of Montréal, in the Haut-Saint-Laurent region of Québec. Depending on the hydric regime, two vegetation types can be recognized: an American elm-black ash (*Ulmus americana* L.-*Fraxinus nigra* Marsh.) community occupying the most hydric sites (approximately 15% of the total area) and a sugar maple-beech community on mesic sites (Brisson, Bergeron & Bouchard, 1992). The present study was conducted in the mesic part of the stand, where drainage varies from moderate to good and slopes are generally less than 5%. The humus is a mull, while the soil is a brown stony loam underlain with surface deposits of morainic origin.

The overstory is dominated by sugar maple, followed in importance by beech, basswood (*Tilia americana* L.), ironwood (*Ostrya virginiana* [Mill.] K. Koch), and hemlock (*Tsuga canadensis* [L.] Carr.). The understory vegetation is typical of a sugar maple-beech community and is characterized by a low percentage of groundcover. Many trees are over 250 years old, with some reaching 300 years of age (Brisson, Bergeron & Bouchard, 1992). Canopy height is approximately 30 m. Canopy gaps were generally small, resulting from branch- and single tree-falls. At present, beech bark disease has had only a very minor effect in the forest. First signs of the disease only appeared around 1990 (Brisson *et al.*, 1996). However, long-term changes in forest composition and dynamics will probably occur in the future as the disease develops (Twery & Patterson, 1984).

The Boisé-des-Muir Ecological Reserve (hereafter referred to as the old-growth stand) was compared to a managed sugar maple stand (hereafter referred to as the managed stand) immediately adjoining the old-growth stand. Both stands are located on the same morainic ridge but differ in their perturbation history; the managed stand has been subjected to at least three partial cuts during the last century (Brisson, Bergeron & Bouchard, 1988). The tree stratum of this stand is dominated by sugar maple, followed by ironwood, bitternut hickory (*Carya cordiformis*, [Wang.] K. Koch) and basswood. American elm, yellow birch (*Betula alleghaniensis* Britton), beech, and hemlock are also present in small numbers. The low number of beech in this stand could be the result of cuts which preferentially eliminated this species (Simard & Bouchard, 1996). Sugar maple seedlings and saplings dominate in the understory.

In the study area, mean daily temperatures range from -10.0°C in January to 20.7°C in July (Environnement Canada, 1982a). Mean annual precipitation totals 960 mm, of which 250 mm fall as snow (Environnement Canada, 1982b).

SAMPLING

In 1993, 44 sampling plots were established in the old-growth stand. The plots were systematically distributed every 10 m along four 100 m parallel line transects which were 20 m apart. Twenty-seven plots were established in a similar fashion in the managed stand (9 plots per transect $\times$ three 80 m transects). Each plot was a series of concentric circles

of different sizes (radius ranging from 0.5 m to 5 m) depending on the variable measured.

SOIL ANALYSIS

In 1993, soils were sampled at the center of each plot. Samples were taken to a depth of 15 cm and were sieved and air dried prior to analysis. Organic matter content was determined by loss on ignition (McKeague, 1976). Exchangeable cations and cation exchange capacity were determined by atomic absorption after extraction in 0.1 M BaCl₂ (Hendershot, Lalande & Duquette, 1993). Total N was determined by wet digestion (Allen, 1989). Available phosphorus was determined upon Bray II extraction (McKeague, 1976). Water content was estimated on subsamples taken in June, July, and August 1993 after drying at 105°C. Litterfall was collected in litter baskets (40 cm × 40 cm) at the center of each plot during autumn (September–November). Leaves were sorted by species, oven-dried, and weighed.

LIGHT MEASUREMENTS

In 1994, light availability was measured above sugar maple and beech seedlings and saplings which had been randomly selected in the sampling plots of the old-growth and managed stands (see sampling method in the “Growth measurements” section below). In 1995, light availability was also measured at four different heights (0.3, 1, 2, and 4 m above the ground) at the center of every plot.

Light availability was determined by measuring the percent of above-canopy photosynthetic photon flux density (400–700 nm, percent PPFD) under completely overcast sky conditions (solar disk not visible, diffuse light only) following the method described in Parent & Messier (1996). Results obtained by Messier & Puttonen (1995) and Parent & Messier (1996) have shown that percent PPFD measured at any time during the day under overcast sky conditions is representative of the mean daily percent PPFD reaching a point in the understory under both clear and overcast conditions. Light measurements were taken using two light sensors (LI-190SA point quantum sensor, LI-COR, Lincoln, Nebraska). One sensor was installed in an open area less than 250 m from the study site and linked to a datalogger (LI-1000, LI-COR, Lincoln, Nebraska) programmed to record every minute the average of 20 readings taken at 5 second intervals. PPFD readings obtained with this sensor were used as estimates of above-canopy PPFD (Q_0). A second light sensor was used to measure PPFD at each sampling point (Q_i), *i.e.*, either above sugar maple or beech individuals, or at different heights at the center of the plots. Percent PPFD was then calculated for each sampling point as:

$$\% \text{ PPFD} = (Q_i / Q_0) \times 100 \quad [1]$$

where Q_i and Q_0 are PPFD values recorded at the same time (± 1 minute). Under completely overcast sky conditions, above-canopy PPFD (Q_0) was quite stable and variation between consecutive minutes was low ($< 0.1\%$).

EMERGENCE AND FIRST SEASON SURVIVAL

In May and September 1993, the number of newly emerged sugar maple and beech seedlings was recorded in each plot (radius = 0.5 m). Survival rates from May to

September were estimated for each species, in each plot, as follows:

$$\% \text{ first season survival} = (n_{\text{September}} / n_{\text{May}}) \times 100 \quad [2]$$

where n_{May} is the number of newly emerged seedlings in May 1993, and $n_{\text{September}}$ is the number of newly emerged seedlings still alive in September of the same year.

GROWTH MEASUREMENTS

Sugar maple and beech individuals of three different size classes were randomly selected in each plot (radius = 5 m) of the old-growth and managed stands. In each plot, one individual of each species and size class was selected, when present. The three size classes were: *i*) 0.3 m to 1 m, *ii*) 1.01 m to 3 m, *iii*) 3.01 m to 5 m. A total of 274 individuals were selected for growth measurements. However, only four beech individuals were found in the plots of the managed stand. Data from those four individuals were not included in the statistical analysis due to the small sample size. At the end of July 1994, annual extension growth of the leader was determined for each individual based on bud scale scars. The leader was defined as the shoot that reached the highest point in the crown. Growth was measured for years 1992 to 1994, and the three-year average was calculated. Since the study site is an ecological reserve, it was not possible to destructively sample saplings in order to obtain diameter growth data.

DISTRIBUTION COEFFICIENTS

The sampling method described in the “Growth measurements” section was also used to determine the distribution coefficients of sugar maple and beech for each of the three size classes described above, in each stand. Distribution coefficients, sometimes referred to as “stocking”, are the percentage of plots in which at least one individual of each species and size class was found.

DENSITY DISTRIBUTION

Density of sugar maple, beech, and other tree species was determined in the old-growth and managed stands by recording the number of individuals of each species (sugar maple, beech, and other species) and size class in each plot. The five size classes were: a) 0.1 m to 0.3 m in height, b) 0.31 m to 1 m in height, c) 1.01 m to 4 m in height, d) height > 4 m and dbh < 5 cm, e) height > 4 m and 5 cm < dbh < 15 cm. The radius of the plots varied with the size class inventoried: for size classes a and b, radius = 2 m; for size class c, radius = 3 m; for size classes d and e, radius = 5 m. Numbers of individuals per m² were converted to numbers per hectare. The origin of each beech individual was recorded as either root sprout or seed. Root sprouts were generally easy to distinguish from individuals of seed origin since root systems were often visible at the surface of the stony soil. It was not possible to excavate the individuals since the study site is an ecological reserve. In case of doubt, the origin was recorded as “unknown”.

STATISTICAL ANALYSIS

For each of the measured variables, comparisons were made between species and/or between stands. For most of the variables, the normality condition could not be met even

after log-, square root- or arcsin-transformation. We therefore used the nonparametric Mann-Whitney test to test for differences between species or between stands. In the few cases where the assumptions for using a parametric test could be met, both the nonparametric Mann-Whitney test and a parametric *t*-test were used. Results for the two tests were consistent and we present only the results of the Mann-Whitney test.

Probability values < 0.05 were considered significant except when multiple tests were done using parts of the same data set (*e.g.*, sugar maple *versus* beech in the old-growth stand, and sugar maple in the old-growth stand *versus* sugar maple in the managed stand). In such cases (*i.e.*, results in Tables III, IV, V, VII, and VIII), we adjusted the significance level in order to account for multiplicity of the tests (P values < 0.025 were considered significant).

Results

SOIL

The total amount of foliar litterfall was not significantly different between the two stands (Table I), but a significantly greater proportion of beech litterfall was measured in the old-growth stand and of other species in the managed stand. Soil humidity was greater in the old-growth stand in June and August, but not in September. The other soil analyses did not generally indicate any significant differences between the two stands, with the exception of exchangeable calcium, magnesium, and cation exchange capacity which were greater in the old-growth stand, and exchangeable aluminium which was greater in the managed stand. While soil properties were relatively similar between stands, soil water content in early summer and exchangeable cations were slightly more favorable to plant growth in the old-growth stand.

TABLE I. Selected soil and litter properties (mean $\pm$ SE) of the old-growth and adjoining managed stands and probability of a significant difference between the stands. Sample size is $n = 44$ for the old-growth stand and $n = 27$ for the managed stand, except for nitrogen (N) for which $n = 6$ for both the old-growth and the managed stands

Variable	Old-growth stand	Managed stand	$P^\dagger$
Amount of litter (g / m ²)	162.0 $\pm$ 5.2	178.9 $\pm$ 7.8	0.090
Litter composition:			
sugar maple (% dry mass)	60.0 $\pm$ 1.7	53.1 $\pm$ 2.7	0.049
beech (% dry mass)	21.2 $\pm$ 1.6	3.1 $\pm$ 0.6	< 0.001
other species (% dry mass)	18.8 $\pm$ 1.7	43.7 $\pm$ 2.7	< 0.001
Humidity in June 1993 (%)	16.4 $\pm$ 0.9	13.1 $\pm$ 0.8	0.015
Humidity in August 1993 (%)	19.4 $\pm$ 1.2	15.9 $\pm$ 1.0	0.025
Humidity in September 1993 (%)	23.1 $\pm$ 0.9	20.9 $\pm$ 1.0	0.110
pH (H ₂ O)	6.32 $\pm$ 0.09	6.09 $\pm$ 0.11	0.150
P available (ug/g)	3.53 $\pm$ 0.79	2.47 $\pm$ 0.28	0.653
Organic matter (%)	18.26 $\pm$ 1.11	15.21 $\pm$ 0.81	0.117
N (mg/g)	6.94 $\pm$ 1.12	5.58 $\pm$ 0.85	0.337
K _e (cmol/kg)	0.265 $\pm$ 0.025	0.236 $\pm$ 0.018	0.507
Ca _e (cmol/kg)	20.905 $\pm$ 2.055	13.374 $\pm$ 1.360	0.005
Mg _e (cmol/kg)	5.85 $\pm$ 0.36	4.03 $\pm$ 0.41	0.003
Al _e (cmol/kg)	0.276 $\pm$ 0.086	0.539 $\pm$ 0.089	< 0.001
Na _e (cmol/kg)	2.96 $\pm$ 0.15	2.57 $\pm$ 0.19	0.137
Mn _e (cmol/kg)	2.36 $\pm$ 0.63	1.57 $\pm$ 0.37	0.295
Cation exchange capacity (cmol/kg)	32.613 $\pm$ 14.350	22.017 $\pm$ 1.752	< 0.001

† Probability associated with the Mann-Whitney *U* value. $p < 0.05$ is considered significant.

LIGHT AVAILABILITY

Light availability (percent PPFD) differed between the old-growth and managed stands at every sampling height from 0.3 to 4 m (Table II). Although percent PPFD was low (generally $< 5\%$) in the understory of both stands, it was slightly but significantly lower in the old-growth stand.

In the old-growth stand, sugar maple seedlings between 0.3 and 3 m in height were located in microsites where percent PPFD was higher than for beech seedlings of similar size (Table III). However, there was no difference between the two species for individuals 3.01 to 5 m in height. There was no significant difference in percent PPFD measured above sugar maple individuals from the old-growth or managed stands (Table III).

EMERGENCE AND FIRST SEASON SURVIVAL

In May 1993, the density of newly emerged sugar maple seedlings was approximately 20 times higher than that of beech in the old-growth stand (Table IV). In September of the same year, the difference in seedling density between the two species was even more pronounced, with only a 15.5% survival rate for beech seedlings compared to 43.0% for sugar maple. Sugar maple seedling density in May and September, and survival rate from May to September, did not differ between the two stands (Table IV).

HEIGHT GROWTH

In the old-growth stand, beech seedlings and saplings 0.3 to 3 m in height had three to four times greater height growth rates than sugar maple individuals of the same size, but the difference was not significant for saplings 3.01 to 5 m in height (Table V). Height growth did not differ between sugar maple from the old-growth and managed stands for any of the three size classes (Table V).

DISTRIBUTION COEFFICIENTS

In the old-growth stand, more than 90% of the plots had beech seedlings (height classes 0.3 to 1 m, and 1.01 to 3 m), and 64% of the plots had at least one beech individual between 3.01 and 5 m high (Table VI). Distribution coefficients of sugar maple were lower than those of beech, especially for 3.01 to 5 m individuals which were present in only 34% of the plots. In the managed stand, however, at least one sugar maple of each size class was found in almost every plot (Table VI). In this stand, distribution coefficients of beech seedlings and saplings were much lower than those of maple, and equal to 0 for the size class 0.3 to 1 m (Table VI).

TABLE II. Light availability (percent PPFD) measured at different heights above the ground at the center of each sampling plot in the old-growth stand ($n = 44$ plots) and in the adjoining managed stand ($n = 27$ plots). Measurements were taken in 1995. Mean $\pm$ SE

Height (m)	Old-growth stand	Managed stand	$P^\dagger$
0.3	2.8 $\pm$ 0.2	4.0 $\pm$ 0.2	< 0.001
1	3.3 $\pm$ 0.3	4.4 $\pm$ 0.2	< 0.001
2	3.6 $\pm$ 0.3	4.7 $\pm$ 0.3	< 0.001
4	4.3 $\pm$ 0.5	5.2 $\pm$ 0.4	0.009

† Probability associated with the Mann-Whitney *U* value. $p < 0.05$ is considered significant.

TABLE III. Light availability (percent PPFD) measured above randomly chosen individuals of sugar maple and beech in the old-growth stand and above sugar maple individuals in the adjoining managed stand. Measurements were taken in 1994. Data for sugar maple individuals from the old-growth stand are compared to those of beech in the old-growth stand (see P_1) and to those of sugar maple in the managed stand (see P_2). Mean $\pm$ SE. Sample sizes are indicated in parentheses

Size class	Old-growth stand: Sugar maple	Old-growth stand: Beech	Managed stand: Sugar maple	$P_1^\dagger$	$P_2^\dagger$
0.3 to 1 m	2.5 $\pm$ 0.2 ($n = 36$)	1.8 $\pm$ 0.1 ($n = 40^\ddagger$)	2.7 $\pm$ 0.2 ($n = 27$)	0.014	0.046
1.01 to 3 m	3.1 $\pm$ 0.3 ($n = 29$)	2.2 $\pm$ 0.2 ($n = 42$)	2.9 $\pm$ 0.1 ($n = 27$)	0.005	0.398
3.01 to 5 m	3.3 $\pm$ 0.3 ($n = 15$)	3.1 $\pm$ 0.2 ($n = 28$)	3.7 $\pm$ 0.3 ($n = 26$)	0.359	0.766

† Probability associated with the Mann-Whitney U value. p values < 0.025 (*i.e.*, $0.05 / 2$) are considered significant since multiple tests ($n = 2$) were done on the same data set.

‡ Of the 40 randomly chosen beech in this size class, 38 were root sprouts and two were of seed origin. We did not distinguish between sprouts and seedlings for the other two size classes.

TABLE IV. Density and survival of newly emerged sugar maple and beech seedlings in the old-growth stand and of newly emerged sugar maple seedlings in the managed stand. Sugar maple data from the old-growth stand are compared to those of beech in the same stand (see P_1) and to those of sugar maple in the adjoining managed stand (see P_2). Mean $\pm$ SE

Variable	Old-growth stand: Sugar maple	Old-growth stand: Beech	Managed stand: Sugar maple	$P_1^\dagger$	$P_2^\dagger$
Density May 1993 (n/ha)	483 497 $\pm$ 57 730	25 188 $\pm$ 7 239	552 017 $\pm$ 75 948	< 0.001	0.455
Density Sept. 1993 (n/ha)	229 878 $\pm$ 31 808	3 184 $\pm$ 1 251	305 732 $\pm$ 49 173	< 0.001	0.215
Survival (%)	43.0 $\pm$ 3.2	15.5 $\pm$ 6.8	48.8 $\pm$ 6.1	< 0.001	0.304

† Probability associated with the Mann-Whitney U value. p values < 0.025 (*i.e.*, $0.05 / 2$) are considered significant since multiple tests ($n = 2$) were done on the same data set.

TABLE V. Height growth (cm/yr) of randomly chosen sugar maple and beech individuals from the old-growth stand and of sugar maple individuals from the managed stand. Height growth is averaged over three years (1992-94). Sugar maple data from the old-growth stand are compared to those of beech in the same stand (see P_1) and to those of sugar maple in the adjoining managed stand (see P_2). Mean $\pm$ SE. Sample sizes are indicated in parentheses

Size class	Old-growth stand: Sugar maple	Old-growth stand: Beech	Managed stand: Sugar maple	$P_1^\dagger$	$P_2^\dagger$
0.3 to 1 m	3.7 $\pm$ 0.4 ($n = 36$)	15.4 $\pm$ 1.3 ($n = 40^\ddagger$)	4.4 $\pm$ 0.6 ($n = 27$)	< 0.001	0.304
1.01 to 3 m	3.8 $\pm$ 0.6 ($n = 29$)	12.2 $\pm$ 1.3 ($n = 42$)	4.7 $\pm$ 0.7 ($n = 27$)	< 0.001	0.389
3.01 to 5 m	3.8 $\pm$ 0.8 ($n = 15$)	9.9 $\pm$ 1.9 ($n = 28$)	4.6 $\pm$ 0.8 ($n = 26$)	0.040	0.665

† Probability associated with the Mann-Whitney U value. p values < 0.025 (*i.e.*, $0.05 / 2$) are considered significant since multiple tests ($n = 2$) were done on the same data set.

‡ Of the 40 randomly chosen beech in this size class, 38 were root sprouts and two were of seed origin. We did not distinguish between sprouts and seedlings for the other two size classes.

DENSITY DISTRIBUTION

Both absolute and relative density of sugar maple regeneration declined markedly with increasing size of the individuals in the old-growth stand (Table VII). In comparison, the absolute density of beech regeneration declined somewhat with increasing size, but the relative density increased (Table VII). Sugar maple regeneration was significantly more abundant than beech in the first two size classes ($0.1 < \text{height} < 1$ m), whereas beech was significantly more abundant than sugar maple for individuals of height > 4 m and dbh < 15 cm (Table VII).

In the old-growth stand, the percentage of beech of seed origin decreased markedly from the 0.1 m to 0.3 m (42%) to the 0.31 m to 1 m (7%) size class (Figure 1). In larger size classes, a higher percentage of the individuals were of unknown origin. Nevertheless, 72% and 59% of beech individuals were identified as root sprouts in the third and fourth size classes, respectively (*i.e.*, $1.01 \text{ m} < \text{height} < 4$ m; and height > 4 m, dbh < 5 cm).

Although absolute density of sugar maple did not differ between the old-growth and managed stands for the first size class (0.1 to 0.3 m), significant differences were observed between the two stands for all other size classes (Table VIII). With increasing size of the individuals, the

TABLE VI. Distribution coefficients (%) of sugar maple and beech individuals of different heights in the old-growth and managed stands. Distribution coefficients are the percentage of plots (radius = 5 m) in which at least one individual of a given species and size class could be found. Old-growth stand ($n = 44$ plots) and managed stand ($n = 27$ plots)

Size class	Old-growth stand		Managed stand	
	Sugar maple	Beech	Sugar maple	Beech
0.3 to 1 m	82	91	100	0
1.01 to 3 m	66	95	100	11
3.01 to 5 m	34	64	96	4

difference between the two stands became more important, and the ratio of sugar maple density in the old-growth *versus* the managed stand was 1:12 for the largest size class (height > 4 m, $5 \text{ cm} < \text{dbh} < 15$ cm); (Table VIII).

Discussion

The causes and patterns of tree replacement or coexistence between sugar maple and beech in old forests of eastern North America remain unexplained despite numerous studies on the subject (Forcier, 1975; Woods, 1979; Cypher & Boucher, 1982; Leak, 1987; Canham, 1989; Runkle,

TABLE VII. Density of sugar maple, beech, and all tree species combined, of different size classes in the old-growth stand. Relative density is presented for sugar maple and beech. The ratio maple:beech is the ratio of sugar maple density to beech density. Mean $\pm$ SE

Size class	Sugar maple density (n/ha)	Sugar maple relative density (%)	Beech density (n/ha)	Beech relative density (%)	Ratio maple:beech	All species density (n/ha)	$P^\dagger$
0.1 to 0.3 m	142 986 $\pm$ 12 754	98.1	1537 $\pm$ 394	1.1	93 : 1	145 717 $\pm$ 12 761	< 0.001
0.31 to 1 m	3 111 $\pm$ 711	68.5	488 $\pm$ 130	10.8	6 : 1	4540 $\pm$ 1063	< 0.001
1.01 to 4 m	434 $\pm$ 115	35.7	482 $\pm$ 104	39.7	1 : 1.1	1214 $\pm$ 171	0.421
height > 4 m, dbh < 5 cm	49 $\pm$ 16	16.4	177 $\pm$ 29	53.4	1 : 3.6	298 $\pm$ 34	< 0.001
height > 4 m, 5 < dbh < 15 cm	26 $\pm$ 9	9.1	188 $\pm$ 25	65.7	1 : 7.2	286 $\pm$ 29	< 0.001

† Probability associated with the Mann-Whitney U value to test difference between sugar maple and beech absolute density. p values < 0.025 (*i.e.*, 0.05 / 2) are considered significant since multiple tests ($n = 2$, see Table VIII) were done on the same data set.

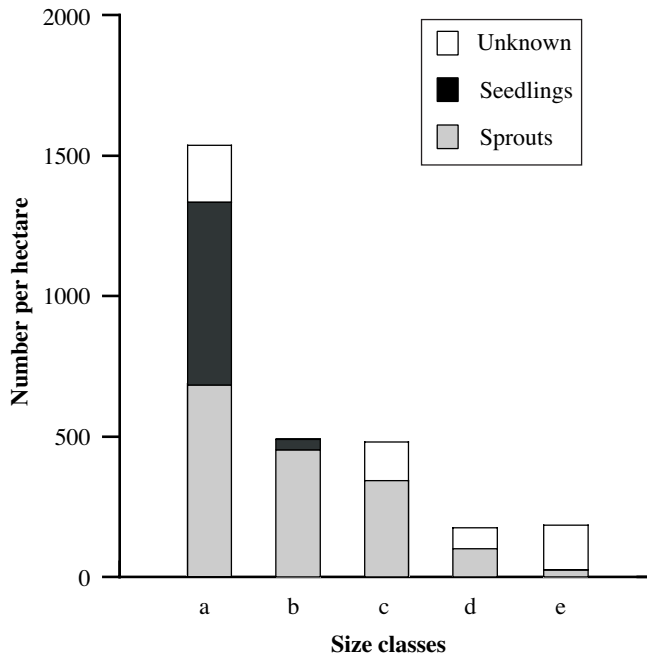


FIGURE 1. Density (number per hectare) of beech individuals of sprout, seed, and unknown origin in each of five size classes in the old-growth stand. Size classes: a) 0.1 m < height < 0.3 m; b) 0.31 m < height < 1 m; c) 1.01 m < height < 4 m; d) height > 4 m and dbh < 5 cm; e) height > 4 m and 5 cm < dbh < 15 cm.

1990; Brisson, Bergeron & Bouchard, 1992; Brisson *et al.*, 1994; Kupfer & Runkle, 1996; Poulson & Platt, 1996). In the present study, data about soils, light regime, seedling emergence and survival, growth, and density distribution of sugar maple and beech regeneration were collected in order to better understand the possible causes for the low number of sugar maple regeneration of sapling size and the gradual replacement of maple by beech in the Bois  -des-Muir old-growth stand.

Several soil characteristics were analyzed to determine if beech litter affected certain soil properties and impaired sugar maple regeneration in the old-growth stand. Our results showed that while soil properties were relatively similar between the old-growth and managed stands, soil water content in early summer and exchangeable cations were slightly more favorable to plant growth in the old-growth stand. The abundant beech litter in the old-growth stand did not act as an efficient physical barrier to sugar maple emergence and establishment since (i) the number of newly emerged sugar maple seedlings in 1993 and their survival through the growing season, and (ii) the number of small sugar maple seedlings (0.1 m to 0.3 m in height) did not differ between the two stands. Therefore, our results do not support the hypothesis that the deficiency in sugar maple regeneration in the sapling size classes in the old-growth stand is due to a negative effect of beech litter and/or soil properties on the early establishment of sugar maple.

Sugar maple density declined more drastically with increasing size in the old-growth than in the managed stand, suggesting that sugar maple mortality was higher in the old-growth stand (Table VIII). Similarly, a comparison of size class distributions between sugar maple and beech in the old-growth stand (Table VII) suggests that sugar maple had a much higher mortality rate than beech in that stand.

Light availability in the understory was evaluated to determine if the relatively low density and possibly higher mortality of sugar maple saplings in the old-growth stand could be related to lower light availability in that stand. Understory light availability (0.3 m to 4 m) was low in both stands (< 5% PPFD), but significantly lower in the old-growth stand (3% to 4% in the old-growth *versus* 4% to 5% PPFD in the managed stand). These differences are not great in absolute terms, but in such a low light environment a difference of only 1% PPFD may be of great biological significance. The lower light conditions observed in the old-growth stand might be related to the presence of beech in

TABLE VIII. Density (n/ha) of sugar maple individuals of different size classes in the old-growth and managed stands. The ratio old:managed is the ratio between sugar maple density in the old-growth stand to sugar maple density in the managed stand. Mean $\pm$ SE

Size class	Old-growth stand	Managed stand	Ratio old:managed	$P^\dagger$
0.1 to 0.3 m	142 986 $\pm$ 12 754	192 460 $\pm$ 25 200	1 : 1.3	0.094
0.31 to 1 m	3 111 $\pm$ 711	9962 $\pm$ 1877	1 : 3.2	< 0.001
1.01 to 4 m	434 $\pm$ 115	2148 $\pm$ 409	1 : 4.9	< 0.001
height > 4 m, dbh < 5 cm	49 $\pm$ 16	424 $\pm$ 57	1 : 8.6	< 0.001
height > 4 m, 5 < dbh < 15 cm	26 $\pm$ 9	321 $\pm$ 57	1 : 12.3	< 0.001

† Probability associated with the Mann-Whitney U value. p values < 0.025 (*i.e.*, 0.05 / 2) are considered significant since multiple tests ($n = 2$, see Table VII) were done on the same data set.

the canopy which casts deeper shade than sugar maple and other temperate tree species (Canham *et al.*, 1994).

Although overall light conditions were lower in the old-growth stand, similar amounts of light were measured above sugar maple individuals in both the old-growth and the managed stands (Table III). Also, sugar maple seedlings and saplings were found in a lower percentage of the plots in the old-growth compared to the managed stand. For instance, sugar maple saplings 3.01 to 5 m in height were present in only 34% of the plots in the old-growth stand, compared to 96% of the plots in the managed stand. These results suggest that (i) the number of microsites where light availability is high enough to sustain sugar maple survival is lower in the old-growth than in the managed stand, and (ii) light availability in the old-growth stand might be near the minimum threshold for sugar maple survival. Other studies conducted in sugar maple stands suggest that the development of sugar maple regeneration > 1 m in height is limited by light availability at approximately 3% PPFD (Messier, 1994).

In the old-growth stand, beech regeneration was located, on average, in more shaded microsites than sugar maple (Table III) but still had a much greater height growth than maple (Table V). Also, the size class distribution of beech in the old-growth stand suggests that beech had a lower mortality than sugar maple (Table VII). Data from a number of different studies suggest that beech is more shade-tolerant than sugar maple (Forcier, 1975; Canham, 1988; 1989; Pacala *et al.*, 1994; Kobe *et al.*, 1995; Poulson & Platt, 1996). Canham (1988) reported greater height growth rate for beech compared to sugar maple under a closed canopy. A lower mortality rate of beech compared to sugar maple has been observed by Kobe *et al.* (1995) in shaded conditions. Kobe *et al.* (1995) suggested that the low mortality of beech might be related to the high proportion of root sprouts among the studied saplings.

Our results also suggest that the cloning habit of beech could be a factor in explaining the ability of this species to grow faster and survive better than sugar maple in deep shade. The number of beech of seed origin declined markedly with increasing size, whereas the decline was less pronounced for beech of sprout origin, suggesting a higher survival of beech sprouts (Figure 1). Sprouts could be advantageous for beech survival in deep shade because (i) they do not have to invest in the construction of a root system, and (ii) they could benefit from subsidies from a well-illuminated parent tree. This might explain why beech individuals up to 3 m in height had greater height growth than adjacent sugar maple individuals (Table V). The advantage of root sprouting for beech growth and survival was experimentally tested in a companion study (Beaudet, unpubl. data). Beaudet found that beech root sprouts 0.1 to 0.3 m in height had much greater height and radial growth rates as well as better survival after experimental defoliation than adjacent beech of seed origin receiving the same amount of light.

The importance of root sprouting for beech varies among sites (Ward, 1961). At Warren Woods (Michigan), Poulson & Platt (1996) found that root sprouting in beech was not very important and that sprouts rarely survived following the death of the parent tree. They concluded that

root sprouting was not a mechanism that favors a shift in dominance towards a beech-dominated community. However, root sprouts that have reached a dbh > 10 cm are frequent at the Boisé-des-Muir. The importance of beech sprouts in this region could be related to the fact that root sprouting tends to increase in importance with increasing latitude (Held, 1983), beech being at the northern part of its native range at the Boisé-des-Muir. Also, the old-growth stand has a relatively stony soil and is located on a morainic ridge. Root systems are frequently exposed at the soil surface, a factor that favors sprout production (Jones & Raynal, 1988). Although beech root sprouts are frequent in some regions and many of their characteristics differ from those of seedlings of seed origin, many studies on beech-maple dynamics have excluded root sprouts from their sampling protocol (Woods, 1979; 1984; Cypher & Boucher, 1982).

In conclusion, while we did not find any evidence for direct or indirect detrimental effects of beech litter on soil properties and early establishment of maple, beech might still be indirectly affecting sugar maple recruitment by casting a very deep shade (Canham *et al.*, 1994). Our results suggest that the relatively low number of sugar maple saplings in the Boisé-des-Muir old-growth stand might be related to a lack of microsites where light availability is high enough to sustain the growth and survival of sugar maple. Beech regeneration seems to be better adapted than sugar maple to survive in the deeply shaded conditions prevailing in the old-growth stand, and we suggest that this is related to its root sprouting ability. We expect that sugar maple will regain its dominance in the understory of the Boisé-des-Muir with increasing stand breakup that is likely to occur in the future. An ongoing study is investigating the hypothesis that a disturbance, such as the severe ice storm that affected the area in January of 1998, could be enough to prevent such forests from maintaining for long periods of time understory light conditions that are detrimental to sugar maple growth and survival.

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Literature cited

- Allen, S. E. (ed.), 1989. Chemical Analysis of Ecological Materials. Blackwell Scientific Publications, Oxford.
- Bouliane, A., 1962. Étude de la transformation de l'érablière à sucre en hêtraie. Thèse M.Sc., Université Laval, Québec.
- Brisson, J., Y. Bergeron & A. Bouchard, 1988. Les successions secondaires sur sites mésiques dans le Haut-Saint-Laurent, Québec, Canada. Canadian Journal of Botany, 66: 1192-1203.
- Brisson, J., Y. Bergeron & A. Bouchard, 1992. The history and tree stratum of an old-growth forest of Haut-Saint-Laurent Region, Quebec. Natural Areas Journal, 12: 3-9.

- Brisson, J., Y. Bergeron, A. Bouchard & A. Leduc, 1994. Beech-maple dynamics in an old-growth forest in southern Québec, Canada. *Écoscience*, 1: 40-46.
- Brisson, J., S. Hauser, É. Beauchamp & A. Bouchard, 1996. État de la maladie corticale du hêtre dans la réserve écologique du Boisé-des-Muir. Rapport de l'Institut de recherche en biologie végétale, Montréal, Québec.
- Canham, C. D., 1988. Growth and canopy architecture of shade-tolerant trees: Response to canopy gaps. *Ecology*, 69: 786-795.
- Canham, C. D., 1989. Different responses to gaps among shade-tolerant tree species. *Ecology*, 70: 548-550.
- Canham, C. D., A. C. Finzi, S. W. Pacala & D. H. Burbank, 1994. Causes and consequences of resource heterogeneity in forests: Interspecific variation in light transmission by canopy trees. *Canadian Journal of Forest Research*, 24: 337-349.
- Côté, B. & J. W. Fyles, 1994. Nutrient concentration and acid-base status of leaf litter of tree species characteristic of the hardwood forest of southern Quebec. *Canadian Journal of Forest Research*, 24: 192-196.
- Cypher, J. & D. H. Boucher, 1982. Beech-maple coexistence and seedling growth rates at Mont Saint Hilaire, Quebec. *Canadian Journal of Botany*, 60: 1279-1281.
- Environnement Canada, 1982a. Normales climatiques au Canada. Volume 2. Température 1951-1980. Environnement Canada, Service de l'environnement atmosphérique, Ottawa, Ontario.
- Environnement Canada, 1982b. Normales climatiques au Canada. Volume 3. Précipitation 1951-1980. Environnement Canada, Service de l'environnement atmosphérique, Ottawa, Ontario.
- Forcier, L. K., 1975. Reproductive strategies and the co-occurrence of climax tree species. *Science*, 189: 808-809.
- Held, M. E., 1983. Pattern of beech regeneration in the east-central United States. *Bulletin of the Torrey Botanical Club*, 110: 55-62.
- Hendershot, W. H., H. Lalande & M. Duquette, 1993. Ion exchange and exchangeable cations. Pages 167-176 in M. R. Carter (ed.). *Soil Sampling and Methods of Analysis*. Lewis Publishers, Boca Raton, Florida.
- Jones, R. H. & D. J. Raynal, 1986. Spatial distribution and development of root sprouts in *Fagus grandifolia* (Fagaceae). *American Journal of Botany*, 73: 1723-1731.
- Jones, R. H. & D. J. Raynal, 1988. Root sprouting in American beech (*Fagus grandifolia*): Effects of root injury, root exposure, and season. *Forest Ecology and Management*, 25: 79-90.
- Kobe, R. K., S. W. Pacala, J. A. Silander Jr. & C. D. Canham, 1995. Juvenile tree survivorship as a component of shade tolerance. *Ecological Applications*, 5: 517-532.
- Kupfer, J. A. & J. R. Runkle, 1996. Early gap successional pathways in a *Fagus-Acer* forest preserve: Pattern and determinants. *Journal of Vegetation Science*, 7: 247-256.
- Leak, W. B., 1987. Fifty years of compositional change in deciduous and coniferous forest types in New Hampshire. *Canadian Journal of Forest Research*, 17: 388-393.
- McKeague, J. A. (ed.), 1976. *Manual of Soil Sampling and Methods of Analysis*. Soil Research Institute, Agriculture Canada, Ottawa, Ontario.
- Melilo, J. M., J. D. Aber & J. M. Muratore, 1982. Nitrogen and lignin control of hardwood leaf litter decomposition dynamics. *Ecology*, 63: 621-626.
- Messier, C., 1994. Relations fonctionnelles entre l'ouverture de la canopée, la lumière et la régénération en sous-couvert : Les cas de l'érablière et de la sapinière. Proceedings of the workshop "La régénération de la zone de la forêt mixte", Ministère des Ressources Naturelles, Québec.
- Messier, C. & P. Puttonen, 1995. Spatial and temporal variation in the light environment of developing Scots pine stands: The basis for a quick and efficient method of characterizing light. *Canadian Journal of Forest Research*, 25: 343-354.
- Pacala, S. W., C. D. Canham, J. A. Silander Jr. & R. K. Kobe, 1994. Sapling growth as a function of resources in a north temperate forest. *Canadian Journal of Forest Research*, 24: 2172-2183.
- Parent, S. & C. Messier, 1996. A simple and efficient method to estimate microsite light availability under a forest canopy. *Canadian Journal of Forest Research*, 26: 151-154.
- Poulson, T. L. & W. J. Platt, 1996. Replacement patterns of beech and sugar maple in Warren Woods, Michigan. *Ecology*, 77: 1234-1253.
- Runkle, J. R., 1990. Gap dynamics in an Ohio *Acer-Fagus* forest and speculations on the geography of disturbance. *Canadian Journal of Forest Research*, 20: 632-641.
- Simard, H. & A. Bouchard, 1996. The precolonial 19th century forest of the Upper St. Lawrence region of Quebec: A record of its exploitation through notary deeds of wood sales. *Canadian Journal of Forest Research*, 26: 1670-1676.
- Tubbs, C. H. & D. R. Houston, 1990. *Fagus grandifolia* Ehrh.-American Beech. Pages 325-332 in R. M. Burns & B. H. Honkala (technical coordinators). *Silvics of North America*. Volume 2 (Hardwoods). U. S. Department of Agriculture, Washington, D.C.
- Twery, M. J. & W. A. Patterson III, 1984. Variations in beech bark disease and its effects on species composition and structure of northern hardwood stands in central New England. *Canadian Journal of Forest Research*, 14: 565-574.
- Ward, R. T., 1961. Some aspects of the regeneration habits of the American beech. *Ecology*, 42: 828-832.
- Woods, K. D., 1979. Reciprocal replacement and the maintenance of codominance in a beech-maple forest. *Oikos*, 33: 31-39.
- Woods, K. D., 1984. Patterns of tree replacement: Canopy effects on understory pattern in hemlock-northern hardwood forests. *Vegetatio*, 56: 87-107.